

Solution Requirements

Requirement Analysis · AI/ML and GenAI Track

Functional requirements

- FR1: Accept name, user ID, age, weight, goal and intensity through a web form.
- FR2: Validate input (age 13–100, weight up to 500 kg, goal and intensity from fixed lists, safe user ID characters).
- FR3: Generate a 7-day workout plan with warm-up, main workout and cooldown/recovery using Gemini.
- FR4: Generate a short nutrition or recovery tip for the chosen goal.
- FR5: Store users, original plan, updated plan and tip in a database.
- FR6: Accept free-text feedback and return a complete revised plan; successive feedback builds on the latest version.
- FR7: Provide an admin page to list and delete users.
- FR8: Show friendly error pages for validation errors, unknown users and AI-service failures.

Non-functional requirements

- Safety: prompts forbid extreme exercise, starvation, dehydration and supplement prescriptions, and advise seeing a professional for pain or medical issues.
- Reliability: retry Gemini calls up to 3 times with exponential backoff on transient errors; 60-second timeout.
- Security: API key and admin credentials kept in environment variables, never in code; user feedback treated as data inside the prompt.
- Usability: simple responsive interface with a loading indicator during generation.
- Portability: runs locally or in Docker with a persistent volume for the database.

Data requirements

Per user: user ID (unique), name, age, weight (kg), goal, intensity, original plan, updated plan, nutrition tip.

Constraints and assumptions

- Requires internet access and a valid Google Gemini API key.
- AI output can vary between runs; it is general wellness guidance, not medical advice.